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Assignment 06

RAG-based Question Answering Bot using ChromaDB + Ollama

1. Objective

The objective of this assignment is to build a Retrieval-Augmented Generation (RAG) based Question Answering system over the Constitution of India using ChromaDB as the vector database and Ollama as the local LLM backend. The system retrieves relevant document chunks and generates accurate answers using a language model.

2. Tools and Technologies Used

- Python
- ChromaDB (Vector Database)
- Ollama (Local LLM Runtime)
- LangChain (for pipeline orchestration)
- Embedding Model (e.g., SentenceTransformers / HuggingFace embeddings)

3. Dataset Description

The dataset consists of the **Constitution of India text document**, which is split into smaller chunks.

- Source: Constitution of India PDF / text file
- Each chunk represents a section or paragraph
- Chunks are embedded and stored in vector database

4. System Architecture (RAG Pipeline)

The system follows a standard RAG pipeline:

1. Document Loading

- Constitution text is loaded from file

2. Text Splitting ○ Large document is split into smaller chunks

3. Embedding Generation ○ Each chunk is converted into vector embeddings

4. Vector Storage ○ Embeddings stored in ChromaDB

5. Retrieval Step ○ User query is converted into embedding ○ Most relevant chunks are retrieved

6. Generation Step ○ Retrieved context is passed to Ollama LLM ○ Final answer is generated

5. Implementation Details

- ChromaDB used for similarity search
- Ollama used for local inference (LLaMA-based model)
- Top-k retrieval strategy applied
- Context window includes retrieved documents + query

6. Evaluation Metrics

To evaluate retrieval performance, the following metrics were used:

6.1 Precision

Measures how many retrieved documents are relevant.

$$\text{Precision} = \frac{\text{Relevant Retrieved Documents}}{\text{Retrieved Documents}} \text{ Total}$$

6.2 Recall

Measures how many relevant documents are successfully retrieved.

$$\text{Recall} = \frac{\text{Relevant Retrieved Documents}}{\text{Relevant Documents}} \text{ Total}$$

6.3 F1 Score

Harmonic mean of precision and recall.

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

7. Results

- Retrieval system successfully fetched relevant constitutional sections for most queries
- Ollama generated coherent and context-aware answers
- Average performance:
 - Precision: Moderate to High ○ Recall: Moderate ○ F1-score: Balanced but dependent on chunk size and embedding quality

8. Failure Case Analysis

8.1 Irrelevant Retrieval

- Some queries retrieved partially unrelated sections
- Reason: Overlapping semantic similarity in embeddings

8.2 Missing Context

- In some cases, retrieved chunks lacked full context of article
- Reason: Chunk size too small

8.3 Hallucination in Generation

- LLM sometimes generated overly generalized answers
- Reason: Insufficient or weak retrieval context

8.4 Query Ambiguity

- Ambiguous questions led to incorrect retrieval results
- Example: vague constitutional references

9. Improvements

- Improve chunking strategy (semantic chunking instead of fixed size)
- Use better embedding models (e.g., BGE, Instructor embeddings)
- Increase top-k retrieval value
- Apply reranking model for better relevance

10. Conclusion

In this assignment, a Retrieval-Augmented Generation system was successfully implemented using ChromaDB and Ollama over the Constitution of India. The system effectively retrieves relevant legal text and generates accurate responses. Evaluation using precision, recall, and F1-score highlights the importance of retrieval quality in RAG systems. Failure analysis shows that improvements in chunking and embeddings can significantly enhance performance.

11. Output:



Constitution of India QA Bot

Ask a question about the Constitution:

Answer

Sure, here is the answer to the question:

Article 1: Name and Territory of the Union

Article 1 establishes that India shall be a Union of States. It defines the territory of the Union and specifies that the States and territories shall be specified in the First Schedule.

```
Retriever.get_relevant_documents was deprecated in langchain-core  
and will be removed in a future release.  
Please use retriever.get_relevant_documents instead.  
docs = retriever.get_relevant_documents(query)  
Precision: 1.0  
Recall: 1.0  
F1 Score: 1.0
```